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A sociology of biocuration: Mapping the landscape

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1. Background

Though there is an emergent literature exploring the history of work relating to biocuration (Leonelli 2016; Stevens 2013; Strasser 2019), and a number of studies that engage with its demographics and working conditions (Burge et al 2012; Holinski et al 2020; Vasilevsky et al 2018; Quaglia et al 2022), there has been **limited sustained qualitative research into the field.**

2. Methodological approach

- Taking an exploratory approach and using snowball sampling, we carried out 10 interviews with individuals working in biocuration (largely, but not exclusively, based in Europe and North America) with the aim of eliciting accounts of the contemporary landscape of biocuration, alongside collating and reviewing relevant literature from the social and natural sciences.
- Interviews were semi-structured (Roulston 2012) and covered topics such as the history and future of biocuration, individuals' trajectories into careers into it, and key challenges facing the field.
- Interviews were transcribed and analysed for themes and patterns.

"Well, one of the things that's needed is … more recognition for the role of the data stewards, you can call them biocurators, you can call them what whatever they are. More recognition for them and the work they do and to understand the value of it because science is, you know, modern science will be built on these databases, so the data better be good." (Interview quote)

4. Reflections

- Biocuration is a key case of contemporary developments in research.
- It demonstrates both the **necessity and semi-invisibility of data work,** as well as the **precarity** of this area of science.
- Its recognition of care and support work and emphasis on collaboration makes it a model of **doing science 'otherwise'.**

"So I think the field of biocuration is very friendly and respectful and helpful. I think that's also something that is hard to quantify … I think there's people who really like helping and those people who really like leading. And I think curators tend to like helping. They want to be part of a bigger project. They want to help the community." (Interview quote)

3. Results

Analysis is ongoing. Here we present some key ways how interviewees described biocuration:

- As a particular set of activities (for instance, working with databases, annotating data, engaging with cutting-edge science in a relevant focal area);
- Interviewees reported diverse – and at times to them unexpected – routes into the field;
- Biocuration was framed as offering more flexible and sometimes more congenial working conditions (particularly in contrast to bench science);
- There were **challenges concerning funding and, more generally, the value of biocuration work being recognised in wider academia;**
- Further challenges included **standardising biocuration practices and languages**, and how to integrate AI and other forms of automation into workflows;
- Some interviewees spoke about personal qualities and characteristics associated with biocuration (such as having an eye for detail);
- Biocuration was also viewed as emphasising **collaboration over competition;**
- Finally, most interviewees described positive affects associated with biocuration: it offered **enjoyment,** as well as allowing for forms of **care** – for data, for science, for others in a community – that were often not valued in other areas of research.

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